Evaluating Privacy Infringement Level in Drone-Captured Images Using Privacy Image Quality Assessment Algorithms

1st Joonyoung Jo
Computer Software
Kunsan National University
Gunsan, Republic of Korea
joon0zo1022@kunsan.ac.kr

2nd Seungmin Park
Computer Engineering
Kyonggi University
Suwon, Republic of Korea
psm0811@kyonggi.ac.kr

3rd Heesung Kim
Software Convergence
Kyung Hee University
Suwon, Republic of Korea
qhrqufdlek@khu.ac.kr

4th Jena Lee
Tourism Development
Kyonggi University
Seoul, Republic of Korea
jena2002@kyonggi.ac.kr

5th Hyeokju Lee
Department of AI and Big Data
Soon Chun Hyang University
Asan, Republic of Korea
hj8678@sch.ac.kr

6th Catherine Morales
Computer and Information Technonolgy
Purdue University
West Lafayette, IN, United States
catherinemorales02@gmail.com

7th Changjun Kim
Cyber Security
Purdue University
West Lafayette, IN, United State
kim4130@purdue.edu

* Eric T. Matson
Computer and Information Technology
Purdue University
West Lafayette, IN, United States
ematson@purdue.edu

***Abstract*—Due to the development of drone technology, drones are increasingly used for observation, monitoring, and capturing. As the technology's usage in public and private spaces increases, the potential for drones to infringe on individual privacy has emerged as a significant social and legal concern. Multiple countries are enacting laws and regulations to prevent privacy violations related to the rapidly changing drone market. However, the clarity of these laws is often insufficient due to the relative nature of privacy definitions.**

This study proposes the criteria to minimize privacy issues during drone captures of the images.

Two central algorithms are applied to define the privacy level. The first is object detection, which makes a bounding box containing a human face or body. The second is the Image Quality Assessment (IQA) algorithm. The input would be an image, and the output would be at a definite privacy level. Privacy Level 0 and Level 1 are defined as whether a part of the body is detected. The level between 2 and 3 is determined by the privacy Image Quality Assessment(PIQA) algorithm. The PIQA algorithm is our main contribution to this thesis.

Since drones can capture images with multiple perspectives, the resolution of the pictures and face directions are primary factors in scoring privacy. Current IQA and Face Image Quality Assessment (FIQA) algorithms have weak correlation problems with resolution and face direction, respectively. By combining FIQA and general IQA, we introduce the known Privacy Image Quality Assessment (PIQA) method, which is robust to face orientation and resolution. By using the object detection method and PIQA, the privacy level between Lv2 and Lv3 is defined.

***Index Terms*—Drone, Object Detection, Privacy Level**

* Corresponding Author

I. Introduction

The drone market has been growing constantly and is expected to be accelerated even further. According to a 2024 survey by market research, the global drone market was valued at $34.5 billion in 2023 and is expected to reach $101.1 billion in 2032 [1]. As it expands the range of the market, drones are being utilized in various fields. On a local scale, Amazon is heavily implementing drone technology to amplify its delivery services to its consumers [2]. Drones can deliver packages faster and more efficiently than traditional methods, reducing delivery times and costs. In addition to their physical laboring capabilities, drone technology carries out dangerous tasks to capture important events and situations in real time. For example, an emergency services task force would be at risk in the affected area and face challenges in developing precise action points. However, specific drone technology excels in conducting perilous inspections within hazardous areas that require accurate real-time scene visuals [3]. Considering the wide range of applications that drone technologies offer, there is potential for more unthinkable utilization in the future.

The rapid growth of drone technology has led to new types of criminal activity, and these criminals are rapidly becoming more sophisticated [4]. Moreover, recent crimes have involved individuals within their homes or properties being filmed without their consent. In 2019, an incident occurred where an unconsenting woman was being filmed in the privacy of her

own home without her consent or awareness [5]. Loretta Alkalay, a former FAA attorney and current aviation law specialist, explains that people only have the right to privacy in spaces where it is explicitly protected, which does not necessarily include their homes or public spaces [6]. This implies that even in personal and private spaces, drones have the right to capture that activity. This challenge further emphasizes the difficulty of demanding privacy protection. These dilemmas highlight the absence of laws that keep pace with innovative technologies and human rights growth. Adequate regulations and management strategies are essential to prevent privacy infringements. These strategies can include universal flight regulations, permitting fly zones, and introducing protective technologies restricting inappropriate drone usage.

Worldwide, legal regulations regarding drone privacy violations are lacking. The Federal Aviation Administration (FAA) focuses primarily on regulations on drone flight safety and does not explicitly address protecting the privacy of individuals [7]. This lack of defined laws for privacy arises because the definition of privacy is relative and can vary depending on the circumstances. Some argue that excessive regulations restrict personal freedoms, while others feel that individual privacy is not adequately protected. This conflicting position arises from the relative nature of the definition of privacy, making it difficult to propose and enforce privacy-related legislation. There is a need for the proposal of regulations that can be widely applied more objectively.

This paper defines privacy concerns as an object detection task, which refers to identifying and classifying objects in digital images or videos. Currently, the accuracy of AI (Artificial Intelligence) in detecting objects surpasses the capability of humans. AI can effectively identify and classify objects, including humans, in drone footage, raising concerns about privacy breaches. As such, establishing criteria from the perspective of AI can address confidentiality issues related to people. The parameters described in this paper that determine the level of privacy include the size of a bounding box and its associated class probability. Several independent variables are considered in this analysis, which include drone altitude, camera orientation, the direction and distance of human detection by the camera, and the object detection model. This research aims to develop recommendations on regulations regarding drone operations to mitigate privacy infringements.

The critical contributions of this work are the following:

1) **New Image Quality Assessment Method.**
The existing IQA, MANIQA, FIQA, and MagFace methods have limitations. IQA shows a weak correlation with face direction distortion, while FIQA shows a weak correlation with resolution distortion. Combined with these two algorithms, we propose a new PIQA (Privacy Image Quality Assessment) method.
2) **Define the privacy level from an AI perspective.**
Previous research quantifying privacy infringement was based on a survey. However, humans distinguish their faces instinctively, and it is hard for humanity to explain how to distinguish them using natural language.
   - Deep learning also has a similar mechanism that is hard to explain.
3) **Drone dataset, which contains relative position differences between humans.**
Few datasets contain the relative pose between humans and drones. We provide our dataset on GitHub.

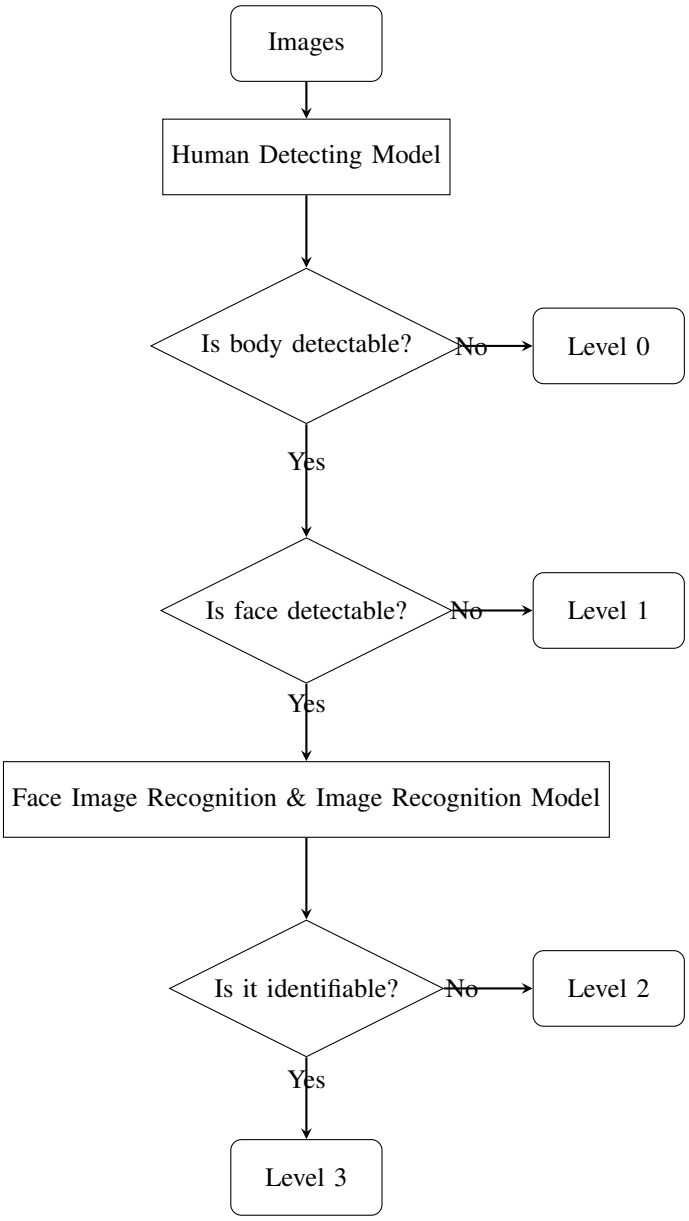


Fig. 1: Flowchart for the privacy level assessment process.

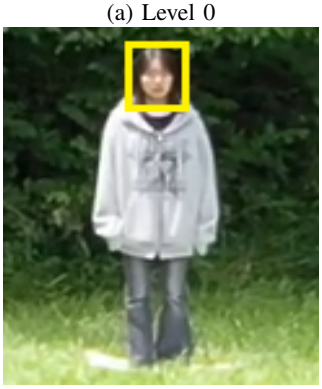
II. Related Work

A. Existing Paper

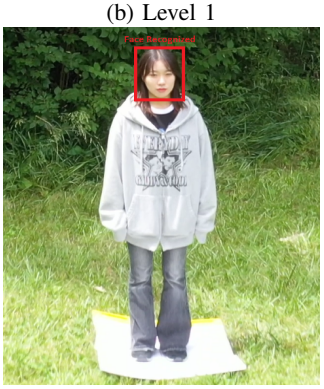
Discussions on privacy norms are actively underway. The following five papers deal with the debate on drone privacy. [8] emphasizes the necessity for privacy protection from two perspectives: privacy related to individuals and confidentiality about legislation. [9] classifies drones by structure and purpose

(a) Level 0

(b) Level 1



(c) Level 2



(d) Level 3

Fig. 2: Face and body detected

and explains how each type of drone can affect personal information protection laws. [10] argued that strengthening the security of drones can infringe on public privacy. Most papers [10], [11] deal with security on the network and drone's cameras. According to the report, unauthorized video taken during drone operations may cause more severe privacy infringement due to security attacks such as Spoofing and Sniffing [11]. It is argued that legal regulations on drone operation and video recording should be implemented to prevent this in advance. However, [12] states that the image forgery attack on the multi-drone network can also be achieved. This paper describes image processing in terms of privacy.

B. Relevant Paper

[13] collected 150 photographs from 15 men and women and applied five different resolutions to each image. Using the image data, a survey of 385 people was conducted, and criteria were developed to evaluate the levels of privacy infringement based on the number of pixels in the image. Based on the images collected by the CCTV simulator, which can adjust the location and angle of the camera, and the privacy infringement criteria, a model for privacy infringement according to the location and angle of the camera was constructed. Instead of using a CCTV simulator, this paper will build data based on images taken by drones in the real world and analyze image data taken from various angles, including front and side photos.

C. Object Detecting model

Object-detection algorithms are composed of two actions: classification and localization. These functions are operated one picture at a time. [14] shows the SOTA algorithm regarding object detection. There are various object detection models, each based on one-stage, two-stage, and the transformer model. [15], [16], [4.4] are examples of the 'one-stage model.' [17], [18], [19] are kinds of 'two-stage model.' Recently, transformer-based object detection has been a stage of art in object detection [20]. In this paper, many object detection algorithms are benchmarked.

D. Face Detection

Face detection is a crucial step in many computer vision tasks, including face recognition and face quality assessment. Various methods have been developed over the years, ranging from traditional approaches to deep learning-based methods. [21] introduced one of the earliest and most widely used face detection algorithms, which uses Haar-like features and a cascade of classifiers. With the advent of deep learning, more robust and accurate methods have emerged. [22] presents a state-of-the-art single-stage detector that uses a feature pyramid network (FPN) to predict face bounding boxes and five facial landmarks. This model achieves high accuracy and speed, making it suitable for real-time applications. [23] is another popular method that employs a multi-task convolutional neural network to detect faces and landmarks simultaneously, ensuring high detection accuracy even in challenging conditions.

E. Image Quality Assessment

Image quality assessment (IQA) aims to evaluate the perceptual quality of images, which can be affected by various factors such as noise, blur, and compression artifacts. Traditional approaches like PSNR and SSIM focus on pixel-wise differences and structural information, respectively [24]. With the advancement of deep learning, data-driven approaches have shown superior performance in IQA tasks. [25] introduces a deep learning-based method that leverages convolutional neural networks (CNNs) to predict image quality scores, trained on large-scale annotated datasets. This method outperforms traditional approaches in capturing complex distortions and perceptual differences in images. Furthermore, [26] proposes a neural image assessment model that predicts aesthetic and technical quality scores using a CNN trained on human-rated images, demonstrating the potential of deep learning in subjective quality assessment.

F. Face Quality Assessment

Face quality assessment (FQA) is a specialized task within IQA, focusing on evaluating the quality of face images for applications like face recognition and verification. The quality of face images can be influenced by factors such as illumination, pose, occlusion, and resolution. [27] introduces an approach for face image quality assessment that predicts the quality score based on the discriminative power of the face

embeddings generated by a face recognition model. Another recent work, [28], proposes a quality-aware face recognition model that incorporates a margin-based loss function to ensure that high-quality images contribute more to the learning process, resulting in more robust and accurate face recognition. These methods highlight the importance of assessing face image quality to improve the performance of downstream face recognition tasks.

III. Definition of Privacy

A. prior researches

Since governments or institutions have not specified protocols to cope with privacy infringement caused by drones, privacy infringement cannot be solved only by law. In addition, since the privacy of individuals is likely to be infringed not only by the use and distribution of captured videos but also by the middle of the capturing, ethical standards must be established to enact legal regulations [29].

To enact legal regulations, the definition of the scope of privacy must precede. However, since confidentiality is defined with different meanings for each individual, problems can arise in trying to draw general conclusions [30].

Despite the ambiguity of privacy, efforts to define its scope are continuing. Westin, a prominent privacy theorist, described it as "an individual, group, or institution deciding for themselves when, how, and to what extent information about them is transmitted to others [30]." According to Steinfeld and Archuleta, privacy means properly using personal information, and determining what is appropriate depends on the situation and the individual's expectations [31].

According to Konvitz and Solove, D.J., opinions about privacy traditionally stem from the difference between "private" and "public," which the individual noted comes from the natural need to distinguish between themselves and the outside world [32]. Of course, the limitations between the private and the public depend on a given era and society, adding that it will create lasting changes throughout the history of what people consider private [33], [34].

In addition, Samuel D. Warren and Louis D. Brandeis, in a famous 1890 article, "The Right to Privacy," were the first to recognize the "threat to privacy caused by technological and social advances" and discussed for the first time the recognition of the right to privacy (defined as the right to neglect) as a general right as a right to guarantee protection against simple emotional distress beyond infringement of property rights [35]. The principle of this right is defined as "an intrusive personality," and the right to neglect stipulates that the protection of unwanted disclosure of private facts, thoughts, feelings, etc. [36], [37] in particular affected the laws of the United States. The provisions became the origin of four privacy tort acts under the U.S. Case Law [38].

B. Current regulations

Legal regulations related to drone privacy around the world are still insufficient. Looking at the legal rules for drone privacy infringement worldwide shows that clear legislation has not yet been prepared in many areas. Article 107 of the Federal Aviation Administration (FAA) focuses on flight safety, but there are limited regulations directly related to the protection of the privacy of others [1]. Many states in the United States have enacted laws about privacy infringement. Indiana State Law Mentioned that the pilot needs to let the state know they will be flying the drone and their intentions and keep their civic pilot duties in mind [2]. California's Civil Code Section 1708.8 prohibits filming videos and audio without the consent of others [39]. These laws have room for subjective interpretation depending on the judge. This paper is expected to be helpful in future trials related to privacy by modeling the privacy level.

C. Definition of Privacy

These days, A.I.'s accuracy in detecting objects surpasses humans' capability. Object detection can lead to an infringement of privacy anywhere, at any time, or in any situation. In this paper, privacy concerns are defined as an object detection task. Setting criteria from the perspective of A.I. can address confidentiality issues related to people.

IV. Methodology

This section proposes a method for extracting optimal-quality outputs from various perspectives. The proposed method integrates the strengths of Image Quality Assessment (IQA) and Face Image Quality Assessment (FIQA) to evaluate both general image quality and face recognition quality. Our approach ensures that the system delivers high-quality performance consistently, regardless of the angle.

A. Detecting Module

1) Body Detecting: To detect a person's body, an object-detecting model called YOLO v4 [40] was used. YOLO v4 is an advanced real-time object detection system with exceptional speed and accuracy.

2) Face Detecting: A face detection module called RetinaFace [22] was used to extract human faces. RetinaFace is a deep-learning-based, cutting-edge facial detector with facial landmarks. RetinaFace is trained using the WIDER FACE dataset [41], including 32,203 images and 393,703 face bounding boxes, with extensive scale, pose, expression, occlusion, and illumination variation. By incorporating facial landmark localization, RetinaFace not only detects faces but also accurately identifies the positions of key facial features such as the eyes, nose, and mouth. This capability improves the model's accuracy in tasks that require precise facial alignment.

B. Privacy Quality Assessment

1) Image Quality Assessment:

a) IQA: The MANIQA (Multi-scale Aggregation Network for Image Quality Assessment) [25] model was employed, assesses overall image quality, considering multiple scales and aggregating features to capture a wide range of quality degradations such as blur, noise, and compression artifacts.

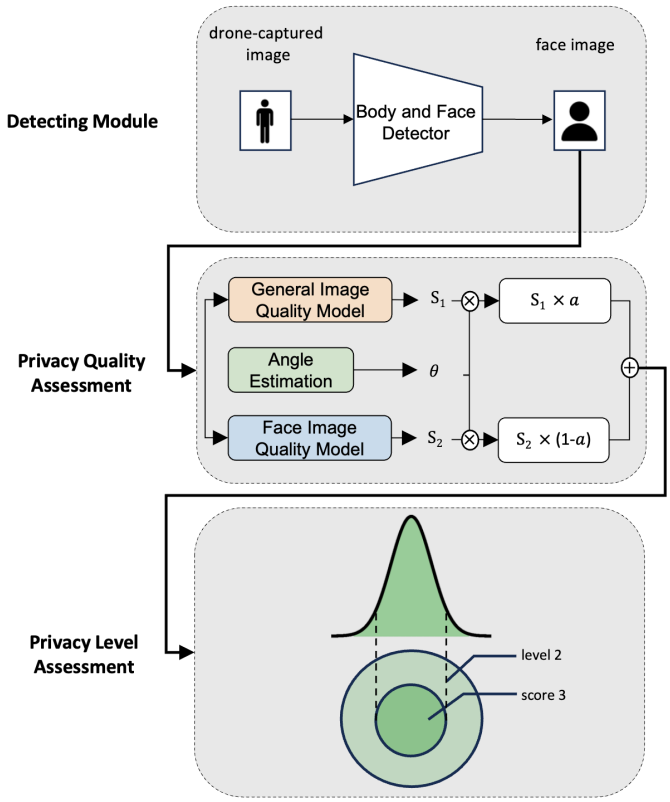


Fig. 3: Overview of the proposed method.

The MANIQA model is based on a multi-scale aggregation network that processes the input image at different scales and combines the results to produce a final quality score. It is Q for an image I is computed as follows:

b) FIQA: MagFace was also employed to assess face-specific quality, integrating face recognition and quality assessment by learning a unified feature representation. It generates a quality score Q based on the magnitude of the face feature vector $\|x\|$:

$$Q = \|x\| \tag{1}$$

Where x is the feature vector of the face image extracted by the MagFace model. The magnitude $\|x\|$ reflects the quality of the face image, with higher magnitudes indicating higher quality.

MagFace's loss function combines ArcFace [42] loss with an additional magnitude-based regularization term. The loss function enhances the learned features' discriminative power while incorporating a quality assessment component. This dual-purpose loss function allows MagFace to simultaneously optimize face recognition accuracy and image quality assessment. It is particularly suitable for applications where both recognition and quality are critical.

The combined loss function L is defined as:

$$L = \frac{1}{N}\sum_{i=1}^{N} L_i \tag{2}$$

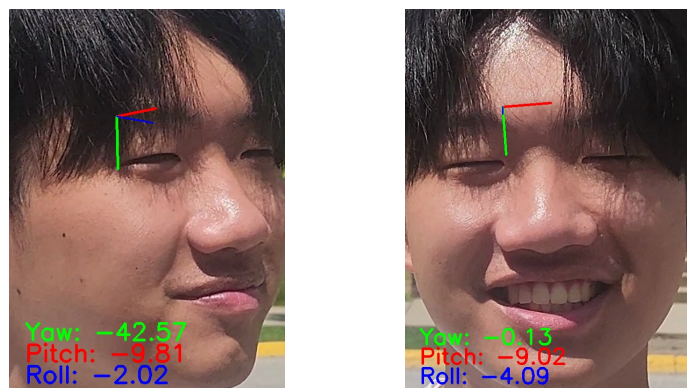
Where L_{softmax} is the softmax loss for face recognition, and L_{quality} is the quality-aware loss that encourages the feature magnitudes to reflect the image quality. The term λ is a weighting factor that balances the two loss components.

The quality-aware loss L_{quality} is defined as:

$$L_i = -\log \frac{e^{s\cos(\theta_{y_i}+m(a_i))}}{e^{s\cos(\theta_{y_i}+m(a_i))} + \sum_{j\neq y_i} e^{s\cos\theta_j}} + \lambda g(a_i) \tag{3}$$

This loss encourages higher-quality images to have larger feature magnitudes, thereby integrating quality assessment directly into the feature learning process.

2) Face Angle Estimator: Hopenet [43] was utilized to estimate a person's face angle, a head pose estimation network trained using the 300W-LP dataset [44], demonstrating strong qualitative performance when evaluated on real-world data. Using Hopenet, it can estimate the orientation of a person's head in three dimensions, providing specific values for yaw, pitch, and roll. In Figure 4a, the person is turned to the side, whereas in Figure 4b, the person is looking directly forward. Hopenet provides precise measurements of yaw, pitch, and roll whenever there is a change in the orientation of a person's head.



(a) A person looking side (b) A person looking front

Fig. 4: Two images looking at the other side.

3) Balancing Mechanism: Faces with significant angle deviations are weighted more heavily by the face image quality model. Conversely, frontal faces are primarily assessed by the general image quality model. The final privacy quality score Q_{privacy} for a face is computed as:

$$Q_{\text{privacy}} = \alpha \cdot Q_{\text{image}} + (1-\alpha)\cdot Q_{\text{face}} \tag{4}$$

Here, α represents the weighting factor that balances the contributions of the general image quality model Q_{image} and the face image quality model Q_{face}. The value of α is determined based on the angles of the face in the image, specifically the roll, pitch, and yaw angles. It is calculated as:

$$\alpha = \frac{\theta_{roll} + \theta_{pitch} + \theta_{yaw}}{3\theta_{max}} \tag{5}$$

C. Privacy Level Assessment

After obtaining the quality scores from the individual models, we normalize and standardize these scores to ensure consistency and comparability. This process involves transforming the scores into a common scale with a mean of 0 and a standard deviation of 1.

First, the scores from the MANIQA and MagFace models are assumed to follow

$$Q_{\text{mani}} \sim N(\mu_{mani}, \sigma^2_{mani}) \Rightarrow Z_{mani} \sim N(0,1)$$
$$Q_{\text{mag}} \sim N(\mu_{mag}, \sigma^2_{mag}) \Rightarrow Z_{mag} \sim N(0,1)$$

where Z_{mani} and Z_{mag} are the standardized scores for the MANIQA and MagFace models, respectively.

$$Q_{\text{privacy}} \sim N(0,1)$$

By standardizing the scores and using a balanced weighting mechanism, our approach ensures that the final privacy [adding].

V. Experiment

This study analyzed how a drone's distance, altitude, and gimbal angle affect the perception of privacy infringement. The DJI Mavic 2 Pro drone was used for this experiment, and a custom Android application was developed using the DJI-SDK [45] (Original SDK [46]). This application provided precise control over the drone's flight, including distance, altitude, and gimbal angle, allowing for accurate adjustments in the drone's positioning and camera orientation.

A. Experimental Set-Up

1) Drone: We used DJI Mavic Pro 2 as a drone. The drone is equipped with a Gymbal camera on the Hasselblad L1D-20c model. The picture taken with this drone camera has a resolution of 20MP and a camera aperture value of f/5.6.

2) DJI-SDK Android Mobile App: We improved the SDK provided by the drone company DJI for the study. Developing Android applications with the SDK heavily depends on Gradle, the build tool integrated with Android Studio. The development environment is Java version 11, Gradle version 7.3.3, and Android Studio version Visual Studio Chipmunk (2021.2.1).

3) Android Mobile Device: The drone was manipulated through an app built by connecting Galaxy Z Flip 4 from Android Studio. The Android version of Galaxy Z Flip 4 is 14.

4) Server Specifications: The experiments were conducted on a server with the following specifications. Intel(R) Core(TM) i9-10940X CPU running at 3.30GHz, 256GB of DDR4 RAM, and an NVIDIA GeForce RTX 3090 with 24GB of memory. The operating system used was Ubuntu 20.04.6 LTS.

5) Software and Libraries: The software environment and libraries used in our experiments are as follows. We used Python 3.8.10 as the programming language, PyTorch 2.3.0 for machine learning tasks, OpenCV 4.9.0 for computer vision tasks, NumPy 1.26.4 for numerical computations, and Matplotlib 3.9.0 for plotting and visualization.

6) Dataset: To evaluate the performance of our algorithms, we utilized the BIWI Head Pose Dataset [47]. The dataset includes depth images and corresponding RGB images, each with a resolution of 640x480 pixels, alongside annotations detailing head pose information. It comprises over 15,000 images and features 24 sequences of 20 individuals (6 females and 14 males, four wearing glasses), recorded while sitting about 1 meter away from the camera. Capturing head poses within a range of approximately ±75 degrees in yaw and ±60 degrees in pitch. Due to its extensive range of head poses, the dataset is a highly regarded resource in the research community.

Given the comprehensive nature of this dataset, we designed our experiments to systematically investigate how variations in drone positioning and camera orientation affect the perception of privacy infringement. The following sections detail the experimental setup and methodology employed in our study.

The resolution R of the image can be defined as the product of the image's width w_{image} and height h_{image}:

$$R = w_{\text{image}} \times h_{\text{image}} \tag{6}$$

The luminance L of a grayscale image can be defined as the average pixel value of the image. For an image with width w_{image} and height h_{image}, the luminance can be calculated as:

$$L = \frac{1}{w_{\text{image}} \times h_{\text{image}}} \sum_{i=1}^{w_{\text{image}}} \sum_{j=1}^{h_{\text{image}}} I(i,j) \tag{7}$$

where $I(i,j)$ represents the pixel value at position (i,j) in the image.

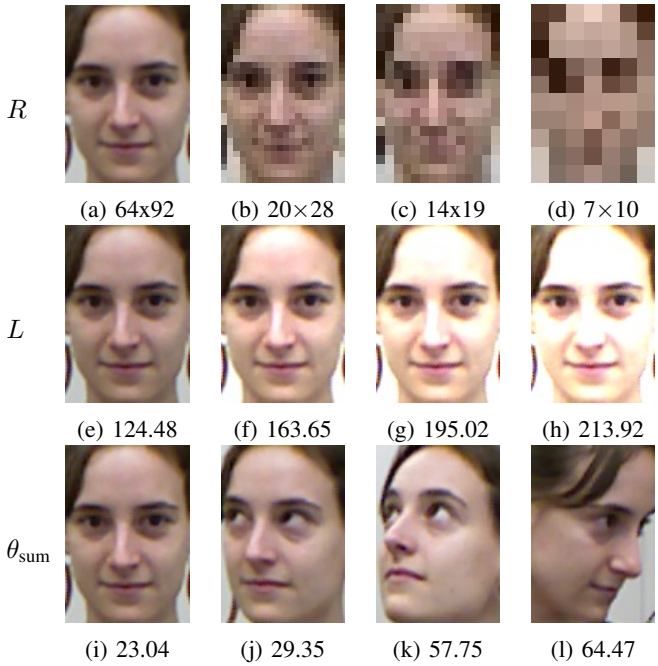
$$\theta_{\text{sum}} = |\theta_{\text{roll}}| + |\theta_{\text{pitch}}| + |\theta_{\text{yaw}}|$$

The sum of these absolute values represents the total angular deviation of the face from a neutral, frontal position.

For instance, a higher θ_{sum} value indicates a greater deviation from a frontal orientation, which can affect the accuracy of face detection and recognition algorithms. Conversely, a lower θ_{sum} value suggests that the face is more aligned with the camera, potentially providing better conditions for accurate image analysis.

In this study, R, L and θ_{sum} are used to measure the privacy quality of image. Lower values of R are considered safer, while higher value of L and θ_{sum} are assumed to indicate a greater level of privacy protection.

By referring to Figure 5, we constructed an inference dataset by varying the values of these three variables R, L and θ_{sum} in increments of 100 within the BIWI Head Pose Dataset.



(a) 64x92 (b) 20×28 (c) 14x19 (d) 7×10

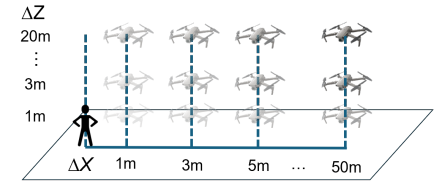
(e) 124.48 (f) 163.65 (g) 195.02 (h) 213.92

(i) 23.04 (j) 29.35 (k) 57.75 (l) 64.47

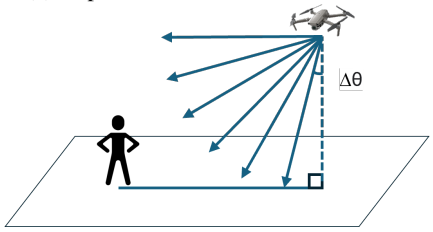
Fig. 5: Value of R, L, θ_{sum} in BIWI Head Pose Dataset

B. Experiment method

Figure 6 shows the location where the drone takes a picture and the experiments' independent variables. Figure 6a was considered to show the relative distance and altitude between the human and the drone. Figure 6b was considered to show the gimbal angle of the drone's camera. Table I provides detailed information on the tested conditions, including Distance ΔX, Altitude ΔY, Gimbal Angle $\Delta\theta$), as presented in Figure 6a and Figure 6b. Each test was conducted under these conditions to ensure a comprehensive evaluation.



(a) Represent the Distance and Altitude



(b) Represent the Gimbal Angle

Fig. 6: Experiment environments.

C. Experimental Result

Figure 7 presents the variations in quality scores based on the sum of angles and image resolution. Figure 7a illustrates how the quality score changes with the sum of angles. As the sum of angles increases, we can observe that the quality scores for MANIQA, MagFace, and our method (PIQA) exhibit different trends, indicating the impact of face orientation on quality assessment. Specifically, the quality scores tend to decrease with increasing angles, highlighting the challenges in accurately assessing quality for non-frontal faces.

Figure 7 shows the quality score variations with respect to image resolution. As the resolution increases, the quality scores for all methods generally improve, demonstrating the influence of image resolution on the perceived quality. Higher resolution images provide more details, which aids in better quality assessment.

As shown in **??**, the correlation between the different quality assessment methods varies significantly, reflecting their respective strengths and limitations in different scenarios. The results underscore the necessity of using a combined approach to achieve more robust and reliable quality evaluations in practical applications [adding].

Table 2 presents the f of the IQA quality score and FIQA quality score model. Image Quality Assessment [adding]

TABLE I: Independent Variables Considered

Variable	Units	Values
Distance ΔX	meters (m)	1, 2, 3, 5, 10, 20, 30, 40, 50
Altitude ΔZ	meters (m)	1, 2, 3, 5, 10, 20, 30, 40, 50
Gimbal Angle $\Delta\theta$	degrees (°)	0, 15, 30, 45, 60, 75, 90

References

[1] Market.us, "Global Drone Market By Type, Technology, End-Industry, and Region - Industry Segment Outlook, Market Assessment, Competition Scenario, Trends, and Forecast 2023-2032," https://market.us/report/drone-market/, 2024, [Online; accessed May-2024].

[2] S. Jung and H. Kim, "Analysis of amazon prime air uav delivery service," *Journal of Knowledge Information Technology and Systems*, vol. 12, no. 2, pp. 253–266, 2017.

[3] M. Lyu, Y. Zhao, C. Huang, and H. Huang, "Unmanned aerial vehicles for search and rescue: A survey," *Remote Sensing*, vol. 15, no. 13, p. 3266, 2023.

[4] Y. Mekdad, A. Aris, L. Babun, A. Fergougui, M. Conti, R. Lazzeretti, and A. Uluagac, "A survey on security and privacy issues of uavs. arxiv 2021," *arXiv preprint arXiv:2109.14442*, 2021.

[5] WINK News. (2022) Collier County woman suspects drone spied on her while she changed. [Online; accessed May-2024]. [Online]. Available: https://winknews.com/2022/09/20/collier-county-woman-spied-on-by-drone-while-changing/

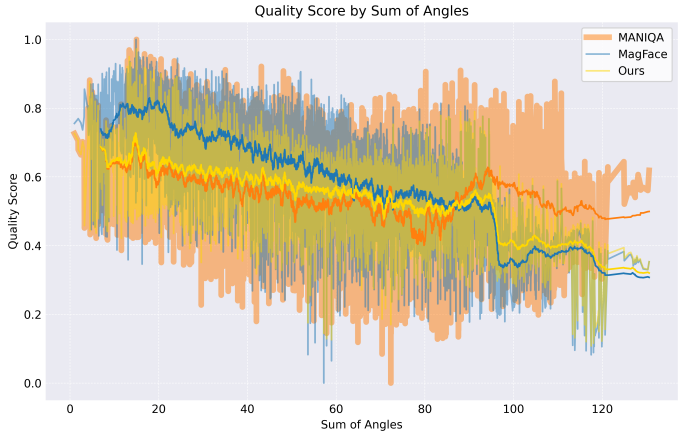
[6] KSLA News. (2024, March) Webster Parish drone operator accused of being a peeping tom. [Online; accessed May-2024]. [Online]. Available: https://www.ksla.com/2024/03/22/webster-parish-drone-operator-accused-being-peeping-tom/

[7] M. Barbee, "Uncharted territory: The faa and the regulation of privacy via rulemaking for domestic drones," *Admin. L. Rev.*, vol. 66, p. 463, 2014.

[8] Y. Mekdad, A. Aris, L. Babun, A. El Fergougui, M. Conti, R. Lazzeretti, and A. S. Uluagac, "A survey on security and privacy issues of uavs," *Computer Networks*, vol. 224, p. 109626, 2023.

[9] P. Quinn and G. Malgieri, "The difficulty of defining sensitive data—the concept of sensitive data in the eu data protection framework," *German Law Journal*, vol. 22, no. 8, pp. 1583–1612, 2021.

[10] B. Nassi, R. Bitton, R. Masuoka, A. Shabtai, and Y. Elovici, "Sok: Security and privacy in the age of commercial drones," in *2021 IEEE symposium on security and privacy (SP)*. IEEE, 2021, pp. 1434–1451.



(a) Privacy Score by Sum of Angles.

(b) Privacy Score by Image Resolution.

Fig. 7: Quality Score Variations Based on Sum of Angles and Image Resolution.

Category	Model	Angle	Tau	Spearman	Pearson
Image Quality Assessment	Brisque [48]	0°	0.61	0.58	0.60
		30°	0.57	0.54	0.56
		60°	0.53	0.50	0.52
		90°	0.49	0.46	0.48
		120°	0.45	0.42	0.44
		150°	0.41	0.38	0.40
		180°	0.37	0.34	0.36
	MANIQA [25]	0°	0.35	0.32	0.34
		30°	0.33	0.30	0.32
		60°	0.31	0.28	0.30
		90°	0.29	0.26	0.28
		120°	0.27	0.24	0.26
		150°	0.25	0.22	0.24
		180°	0.23	0.20	0.22
Face Image Quality Assessment	SDD-FIQA [18]	0°	0.43	0.42	0.44
		30°	0.40	0.39	0.41
		60°	0.38	0.37	0.39
		90°	0.36	0.35	0.37
		120°	0.34	0.33	0.35
		150°	0.32	0.31	0.33
		180°	0.30	0.29	0.31
	MagFace [28]	0°	0.57	0.55	0.56
		30°	0.54	0.52	0.53
		60°	0.51	0.49	0.50
		90°	0.49	0.47	0.48
		120°	0.47	0.45	0.46
		150°	0.45	0.43	0.44
		180°	0.43	0.41	0.42
Privacy Quality Assessment	SDD-FIQA [18]	0°	0.45	0.44	0.46
		30°	0.42	0.41	0.43
		60°	0.40	0.39	0.41
		90°	0.38	0.37	0.39
		120°	0.36	0.35	0.37
		150°	0.34	0.33	0.35
		180°	0.32	0.31	0.33
	MagFace [28]	0°	0.57	0.56	0.55
		30°	0.54	0.53	0.52
		60°	0.51	0.50	0.49
		90°	0.49	0.48	0.47
		120°	0.47	0.46	0.45
		150°	0.45	0.44	0.43
		180°	0.43	0.42	0.41

TABLE II: Comparison of Correlation Coefficients for Three Types of Models Across Different Angles

TABLE III: Independent Variables Considered

Method	Tau	Va
Mean	0.72	0.70
Angle-base	0.74	0.78
Sigmoid-angle	0.94	0.43

[11] H. J. Hadi, Y. Cao, K. U. Nisa, A. M. Jamil, and Q. Ni, "A comprehensive survey on security, privacy issues and emerging defence technologies for uavs," *Journal of Network and Computer Applications*, vol. 213, p. 103607, 2023.

[12] M. Yahuza, M. Y. I. Idris, I. B. Ahmedy, A. W. A. Wahab, T. Nandy, N. M. Noor, and A. Bala, "Internet of drones security and privacy issues: Taxonomy and open challenges," *IEEE Access*, vol. 9, pp. 57 243–57 270, 2021.

[13] S.-h. Park and K.-h. Lee, "Developing criteria for invasion of privacy by personal drone," in *2017 International Conference on Platform Technology and Service (PlatCon)*. IEEE, 2017, pp. 1–7.

[14] S. S. A. Zaidi, M. S. Ansari, A. Aslam, N. Kanwal, M. Asghar, and B. Lee, "A survey of modern deep learning based object detection models," *Digital Signal Processing*, vol. 126, p. 103514, 2022.

[15] R. Girshick, "Fast r-cnn," in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 1440–1448.

[16] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 2117–2125.

[17] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 779–788.

[18] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "Ssd: Single shot multibox detector," in *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14*. Springer, 2016, pp. 21–37.

[19] Y. Sun, L. Dong, S. Huang, S. Ma, Y. Xia, J. Xue, J. Wang, and F. Wei, "Retentive network: A successor to transformer for large language models," *arXiv preprint arXiv:2307.08621*, 2023.

[20] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *European conference on computer vision*. Springer, 2020, pp. 213–229.

[21] P. Viola and M. Jones, "Robust real-time object detection," *International Journal of Computer Vision*, vol. 57, no. 2, pp. 137–154, 2004.

[22] E. V. I. K. S. Z. Jiankang Deng, Jia Guo, "Retinaface: Single-stage dense face localisation in the wild," *arXiv preprint arXiv:1905.00641*, 2019.

[23] Z. L. Y. Q. Kaipeng Zhang, Zhanpeng Zhang, "Joint face detection and alignment using multi-task cascaded convolutional networks," *IEEE Signal Processing Letters*, vol. 23, no. 10, pp. 1499–1503, 2016.
[24] H. R. S. E. P. S. Zhou Wang, Alan C. Bovik, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.
[25] M. QA, "Deep learning-based image quality assessment," *Journal Name*, vol. Volume, no. Number, p. Pages, Year.
[26] H. Talebi and P. Milanfar, "Nima: Neural image assessment," *IEEE Transactions on Image Processing*, vol. 27, no. 8, pp. 3998–4011, 2018.
[27] J. Y. D. Y. S. Z. L. Xiang Zhu, Zhen Lei, "High-fidelity pose and expression normalization for face recognition in the wild," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 37, no. 6, pp. 1224–1236, 2015.
[28] N. L. S. Z. L. Xiang An, Zhen Lei, "Magface: A universal representation for face recognition and quality assessment," *arXiv preprint arXiv:2103.06627*, 2021.
[29] A. Cavoukian, *Privacy and drones: Unmanned aerial vehicles*. Information and Privacy Commissioner of Ontario, Canada Ontario, 2012.
[30] J. A. Jacko, "Human computer interaction handbook: Fundamentals, evolving technologies, and emerging applications," 2012.
[31] R. M. Thompson and M. Richard, *Domestic Drones and Privacy: a primer*. Congressional Research Service Washington, DC, 2015, vol. 43965.
[32] L. Steinfeld and K. S. Archuleta, "Privacy protection and compliance in higher education: The role of the cpo." *Educause Review*, vol. 41, no. 5, p. 62, 2006.
[33] M. R. Konvitz, "Privacy and the law: A philosophical prelude," *Law and Contemporary Problems*, vol. 31, no. 2, pp. 272–280, 1966.
[34] M. D. Szabó, "Kísérlet a privacy fogalmának meghatározására a magyar jogrendszer fogalmaival," *Információs Társadalom: társadalomtudományi folyóirat*, vol. 5, no. 2, pp. 44–54, 2005.
[35] D. J. Solove, "Conceptualizing privacy," *Calif. L. Rev.*, vol. 90, p. 1087, 2002.
[36] I. R. Kramer, "The birth of privacy law: A century since warren and brandeis," *Cath. UL Rev.*, vol. 39, p. 703, 1989.
[37] M. E. Bratman, "Hierarchy, circularity, and double reduction," 2002.
[38] R. A. Posner, "The right of privacy," *Ga. L. Rev.*, vol. 12, p. 393, 1977.
[39] California Legislature, "Assembly Bill No. 856: An act to amend Section 1708.8 of the Civil Code, relating to privacy," California Legislative Information, 2016, [Online; accessed Month-Year]. [Online]. Available: https://leginfo.legislature.ca.gov/faces/billTextClient.xhtml?bill_id=201520160AB856
[40] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "Yolov4: Optimal speed and accuracy of object detection," 2020.
[41] S. Yang, P. Luo, C. C. Loy, and X. Tang, "Wider face: A face detection benchmark," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 5525–5533.
[42] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "Arcface: Additive angular margin loss for deep face recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2019.
[43] N. Ruiz, E. Chong, and J. M. Rehg, "Fine-grained head pose estimation without keypoints," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, June 2018.
[44] X. Zhu, Z. Lei, X. Liu, H. Shi, and S. Z. Li, "Face alignment across large poses: A 3d solution," *CoRR*, vol. abs/1511.07212, 2015. [Online]. Available: http://arxiv.org/abs/1511.07212
[45] "Refined DJI-SDK for DJI Mavic Pro 2," https://github.com/K-SW-SLAMDUNK/anony-drone/tree/main/SDK/Mobile-SDK-Android/, 2024.
[46] DJI, "DJI Mobile SDK V4," https://github.com/dji-sdk/Mobile-SDK-Android, 2023.
[47] G. Fanelli, M. Dantone, J. Gall, A. Fossati, and L. Van Gool, "Random forests for real time 3d face analysis," *Int. J. Comput. Vision*, vol. 101, no. 3, pp. 437–458, February 2013.
[48] A. Mittal, A. K. Moorthy, and A. C. Bovik, "No-reference image quality assessment in the spatial domain," *IEEE Transactions on Image Processing*, vol. 21, no. 12, pp. 4695–4708, 2012.

APPENDIX A
APPENDIX